

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Theory of Computation**

Code No: **CSC262** Semester: **4th Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. List the application of Pumping Lemma. Minimize the following DFA using Table Filling Algorithm.
2. Define yield in parse tree with example. Convert the following grammar to CNF $(A \rightarrow ASB \mid \epsilon)(A \rightarrow 0AS \mid 0)(B \rightarrow S \mid S \mid A \mid 1 \mid 1)(S \rightarrow \epsilon)$
3. Distinguish between multi - tape and multi - track Turing Machine. Design a TM that accepts a string over alphabet $\Sigma = \{a, b\}$, such that $L = \{a^n b^n \mid n > 0\}$.

Section B

Attempt any Eight questions[8×5=40]

4. Why do we need asymptotic notation? Discuss about alphabet and power of alphabet.
5. Define ambiguous grammar. How do you use parse tree to show the ambiguity of a grammar?
6. Given the non-deterministic Turing machine, $(TM: (\{q_0, q_1, q_2, q_3\}, \{0, 1\}, \{0, 1, B\}, \delta, q_0, B, \{q_3\}))$ with the following transition rules, describe the language accepted by given Turing machine. $\delta(q_0, 0) = \{(q_0, 1, R), (q_1, 1, R)\}; \delta(q_1, 1) = \{(q_2, 0, L)\}; \delta(q_2, 1) = \{(q_0, 1, R)\}; \delta(q_1, B) = \{(q_3, B, R)\}$
7. Given two DFAs (M_1) and (M_2) over the same alphabet Σ , design a third DFA (M_3) such that (M_3) will accept the string $w \in \Sigma^*$ if it is accepted by both (M_1) and (M_2) .
8. Differentiate between Moore and Mealy machines.
9. Design a DFA that accepts the odd length of binary strings
10. How do you convert CFG to PDA? Explain.
11. Why do we need ϵ transition? Give the formal definition of PDA.
12. Describe about abstract, decision and optimization problems.